

PERFORMANCE CHARACTERISTICS TEST ON PLUNGER PUMP

AIM

To find out the efficiency of the plunger pump and plot the following graphs.

1. Discharge vs efficiency.
2. Discharge vs head.
3. Discharge vs Input power.

PRINCIPLE

$$\text{Efficiency of the pump, } \eta_p = \frac{\text{Power Output}}{\text{Power Input}} \times 100 \%$$

$$\text{Power Input, IP} = \frac{n \cdot 3600 \cdot \eta_m \cdot \eta_T}{T \cdot E_m} \text{ kW}$$

Where,

n = Number of revolution of energy meter (10 Rev)

T = Time taken for 10 revolution in sec.

E_m = Energy meter constant Rev/kW hour.

η_m = Efficiency of motor = 80 %.

η_T = Transmissions efficiency = 90 %

$$\text{Output power, OP} = \frac{\rho \cdot g \cdot Q_{act} \cdot H}{1000} \text{ kW}$$

Where,

ρ = Density of water = 1000 kg/m³
 g = acceleration due to gravity in m/s² (9.81 m/s²)

Q_{act} = Actual discharge in m³/sec. = $\frac{A \cdot R}{t}$ m³/s

Where,

A = Area of collecting in m² = (0.6 x 0.6) m²

OBSERVATION AND TABULAR COLUMN.

[illegible]

- R = Rise of water in m (say 0.05m) in collecting tank.
 t = Time in sec for the rise of 0.05 m of water in the tank.
 H = Total head in m

Total Head, $H = \text{Suction head}(h_s) + \text{Delivery head}(h_d) + \text{Datum head}(h_z)$

Coefficient of discharge, $C_d = \frac{Q_{act}}{Q_{th}} \quad \text{m}^3/\text{s}$

Theoretical discharge, $Q_{th} = \frac{L a N}{60} \quad \text{m}^3/\text{s}$

Where,

L = length of stroke in m (0.052 m)

a = Area of cross section of piston in $\text{m}^2 = \frac{\pi}{4} d^2$ (d = diameter of piston) = (0.030 m)

N = speed of pump in rpm.

Percentage of Slip = $(1 - C_d) \times 100$

PROCEDURE

1. Open the delivery valve.
2. Start the pump and allow the flow of water to stabilize.
3. Adjust the delivery valve to bring the delivery pressure to 10 kg/cm².
4. Note the reading of the suction pressure gauge.
5. Note the time 't' taken to rise of water level 5 cm in the collecting tank.
6. Note the time 'T' taken for 10 revolution of energy meter disc.
7. Measure the speed of the crank.
8. Repeat this for delivery pressure readings 8 Kg/cm², 6, 4, 2 Kg/cm².
9. Switch off the pump.

GRAPHS

1. Discharge v/s efficiency.
2. Discharge v/s head.
3. Discharge v/s Input power.

RESULT

INFERENCE